Journal of Agricultural Science; Vol. 18, No. 5; 2026
ISSN 1916-9752 E-ISSN 1916-9760
Published by Canadian Center of Science and Education

Costs and Benefits of Wetland Retention in the Riding Mountain Agricultural Region of Manitoba

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Received: February 12, 2026 Accepted: March 15, 2026 Online Published: April 15, 2026

doi:10.5539/jas.v18n5p1 URL: https://doi.org/10.5539/jas.v18n5p1

Abstract

Canada's agrifood sector is significant within the economy, contributing over 2.3 million jobs along with $144 billion annually in GDP. Expansion of the farmland base since European settlement has provided an abundance of land, but there are concerns about the ecological cost of continued expansion. Using a wetland conservation cost model with spatially explicit dimensions, this research estimates the economic benefits of wetland drainage in a sub-basin of the Assiniboine West Watershed District in Manitoba, concluding that soil productivity and commodity prices significantly affect decisions to drain wetlands. Options for optimal policy based upon specific ecosystem value are outlined, and policy costs are estimated.

Keywords: cost benefit analysis, conservation costs, net present benefit, policy design, ecological value, spatial mapping

1. Introduction

Canada's agriculture and food sector accounts for approximately 7% of its GDP (around $144 billion), and in addition to providing food for citizens both nationally and internationally, the industry is a significant source of employment, contributing over 2.3 million jobs (AAFC, 2026). There are, however, costs associated with increased production of commodity and food products, as expansion of the base of agricultural land can contribute to degradation of the environment and local ecosystems. In particular, the draining of wetlands – generally done to increase the usable acreage of a parcel of farmland thereby increasing the size of a farm's production and revenue – is one of the primary drivers of this environmental degradation. According to some estimates, over 40% of Canadian Prairie wetlands have disappeared since settlement, with agricultural development being a major contributor to this loss (Environment Canada, 1986; Watmough & Schmoll, 2007). There may be a discrepancy between the wetland conservation costs, for which farmland owners are responsible, and benefits to the public which are generated through conservation activities, as these are shared more broadly among citizens. Determining the societal value of retained wetlands presents a problem since it is challengimg to quantify the monetary benefits of wetlands.

Current policies in various regions in North America have not always adequately protected wetlands in areas of intensive agricultural production. Policies designed to reduce the rate of wetland drainage on agricultural land are important to ensure that environmental and agricultural considerations are both taken into account. These policies also have the potential to affect farm income, and accordingly farmland prices and by extension, farm wealth. Accordingly, any policies affecting farm income may also affect the enterprise's long-term financial health. Because significant portions of farmland are passed from parents to children, there are also inter-generational wealth impacts of policies that affect farmland values.

The objective of the research reported in this paper is to determine the economic benefits for private landowners of wetland drainage in a sub-basin within the Assiniboine West Watershed District in Manitoba. A secondary contribution of this research is to provide a basis for development of a wetland supply curve for the study region. Calculation of the public value of wetland retention will use estimates carried out in previous work; these estimates will facilitate a cost-benefit analysis of potential wetland conservation policies that minimize the

environmental impact in a way that is consistent with agri-food sector growth. Lastly, this research builds upon findings of previous researchers and adds to the literature by studying a region which has to this point received comparatively little attention.

This research also incorporates theories related to government intervention in private markets when negative externalities occur as the result of market failure. Existing literature explores the importance of designing policies to address concerns over private landowners draining too many wetlands. Inherently, the type of benefits provided by wetlands to society are public goods, however, they are often located on private lands. This conflict can lead to outcomes where more than the socially optimal number of wetlands are drained if there is no government intervention because private landowners use the lands to increase their agricultural production and income (Heimlich, 1994). This situation can lead to the development of policies intended to compensate landowners for *not* draining wetlands. Ideally, the level of compensation will match the overall value to society that the retained wetlands provide (Boardman et al, 2018). A major challenge with respect to policy design is then determining the societal value of the retained wetlands, as outlined in greater detail below.

The remainder of this paper is divided into four main parts. The next section provides a review of previous work in this area that provides a foundation for the empirical approach undertaken in this paper. After that, an overview of the data and empirical methodology used to estimate the private and public benefits of wetland retention and drainage is presented. The next-to-last section of the paper presents estimation results for the various scenarios along with a discussion of the policy implications of the results, and the paper concludes with a summary of main findings, acknowledgement of limitations and suggestions for future research.

2. Previous Research

Much of the previous related literature focussed upon the factors which influence agricultural landowners' land use decision making, particularly with respect to decisions of whether to drain a wetland for agricultural purposes. Drainage of wetland typically involves removal of surface water from land that could be productive in agricultural production via use of ditches and other techniques. Encouraging private landowners to conserve wetlands thus requires compensation that roughly equals the opportunity cost of the agricultural land, less drainage costs. Results of research by Heimlich (1994) bolsters the conclusion that forgone farm profitability from converting wetlands to agricultural use could serve as a substitute for costs associated with private marginal wetland conservation costs.

Multiple previous studies have carried out estimates of wetland conservation costs; a portion of this work has been founded upon calculation of prices of permanent easements. Easements place restrictions upon the development and of wetland basins, implying reduced land values reflect forgone agricultural production. Shulz and Taff (2004) used a hedonic regression method to find that each additional acre of eased permanent wetland reduced sales prices by an average of $67/ha in the southeastern part of the state of North Dakota. Work by Lawley and Towe (2014) that used a propensity matching methodology estimated wetlands under permanent easement caused a reduction in farmland values of $37/ha in Manitoba, and found easement programs have successfully conserved at-risk habitat. Their results indicate that landowners have seen land values decline for the average parcel of land with easements. The authors also found that more productive land parcels have fewer easements compared to less productive land, suggesting landowners are entering easement agreements for parcels of land with less value for agriculture. Accordingly, there may be a limit on the ability of easement programs to achieve wetland retention at scale. Hill et al. (2011) used a discriminative reverse auction procedure to estimate an average restoration cost for drained wetland in Saskatchewan's Assiniboine River Watershed at approximately $294/ha annually over a 12-year contract.

De Laporte (2014) explored the effects of crop prices, wetland regulation and nuisance costs in Saskatchewan, finding that the price of agricultural products was the most important factor in determining the total acreage of wetlands that would be drained. In periods of lower agricultural commodity prices, regulation was found to play a less critical role in preventing drainage of wetlands; by comparison when prices are high, regulation was found to be too expensive to incentivize landowners to retain wetlands. Gelso, Fox, and Peterson (2008) explored the difference between certainty equivalence from farming cropland with no wetlands compared to cropping a field that has wetlands using a contingent valuation method and employing data from agricultural producers in Kansas. They discovered the presence of wetlands has a negative impact upon profits as a result of nuisance costs. The authors also found that perceived nuisance costs increase as the wetland area increases and become even higher when smaller wetlands are dispersed throughout the agricultural land, and that this has implications for effective targeting of wetland policies.

Some studies have attempted to look more broadly at wetland retention rather than just focussing upon costs. Cortus et al. (2011) hypothesized representative farm consistent with attributes from a rural municipality in the east-central region of Saskatchewan and used a dynamic stochastic simulation model of farm cash flows to estimate wetland drainage costs and benefits. That work found that public benefits of wetland retention in the study area exceed annual average private costs, concluding that current NGO and government cost-sharing programs and other payments fail to adequately incentivize permanent wetland retention.

The Canadian province of Alberta has been the subject of multiple studies exploring the economics of wetland conversion. Clare et al. (2021) found economic returns to farmers from wetland drainage to be variable, with producers experiencing losses from drainage under some circumstances. Using a net present value framework, Asare, Lloyd-Smith, and Belcher (2022) modeled economic benefit to farmers of wetland drainage in Alberta, to calculate a supply curve for wetlands, showing private benefits of draining wetlands demonstrate significant variation within a watershed. That research showed wetland conservation strategies are more effective when targeted, compared to a blanket conservation policy based on the assumption all wetlands have a homogeneous set of costs and benefits. The authors accomplished this by developing a set of four conservation policy targeting scenarios for wetlands, based on the Alberta government's classification system which rates every wetland using a four-level scale. Using this scale, the authors developed four scenarios. The first was a high benefit targeting scenario which had the goal of conserving wetlands possessing comparatively high ecosystem functions. The second was a targeting scenario aimed at minimizing costs associated with wetland conservation through relatively low cost of conservation. The third scenario targeted low benefits through focusing wetlands with lower functions associated with the ecosystem, and the fourth and final scenario attempted to randomly conserve approximately half of all wetlands in the study area. The authors ranked conservation costs for individual wetlands from lowest to highest to calculate a supply curve for wetlands, allowing them to explore the varying levels of conservation costs required for each policy targeting scenario to determine the most efficient. The research reported in this paper undertakes a similar analysis by constructing three hypothetical policies based on the results of previous research regarding the heterogeneity of wetland ecosystem benefits.

This paper follows the approach of Asare, Lloyd-Smith, and Belcher (2022) outlined above to determine heterogeneity of wetland drainage costs by utilizing GIS software to measure the distance of wetlands to the nearest watercourses. This distance proxy measures the cost to a private landowner of draining a wetland. The closer a wetland is to the nearest watercourse, the cheaper it will be for the landowner to drain it because it would require less time and effort spent on the excavation due to the shorter distance and potentially shallower ditches required. This drainage cost variable is then employed in the net present benefit equation, which introduces heterogeneity in conservation costs to the model to simulate reality more closely.

3. Data and Procedure

The study area for this research is a sub-basin within the Assiniboine West Watershed District in the province of Manitoba, covering 1,240 km^2 and containing 2,549 wetlands (Figures 1 and 2). This area was chosen because of the considerable number of wetlands located within private agricultural lands and the importance of these wetlands due to their proximity to Riding Mountain National Park, one of only two national parks in Manitoba and the only that is easily accessible (the other, Wapusk National Park, is located in extreme northeastern Manitoba). No previous estimates of the value of wetland conservation have been carried out for this agricultural region.

The study area lies within North America's Prairie Pothole Region (PPR), one of the most ecologically significant wetland landscapes in the world. The PPR spans portions of Alberta, Saskatchewan, Manitoba, as well as the northern United States and contains millions of shallow depressional wetlands formed by glacial activity. These wetlands provide an array of ecosystem services including water storage, flood mitigation, nutrient retention, sequestration of carbon, and habitat for migratory waterfowl and other wildlife. Because of the large number and spatial distribution of wetlands across agricultural landscapes, the PPR has become a focal region for research examining the ecological and economic consequences of wetland drainage and conservation policies (Gleason et al., 2011; Ross & McKenna, 2023). Wetland drainage in this (or any) region is characterized by tradeoffs between private and public benefits and costs, and therefore particularly important for both environmental management and agricultural policy.

Wetlands in the Canadian Prairie Provinces, including Manitoba, have experienced substantial losses due to agricultural development and drainage practices over the past century. Expansion of crop production and improvements in drainage technology have encouraged landowners to convert wetlands into arable land, resulting in widespread alteration of prairie landscapes. Numerous studies have documented the ecological

consequences of this process, including reductions in biodiversity, increased nutrient runoff, and diminished flood storage capacity (Whitfield et al., 2024; Bartzen et al., 2010). Because agriculture remains the dominant land use across much of Manitoba, wetlands located on privately owned farmland are especially vulnerable to drainage. As a result, the region represents an important setting in which to examine the economic incentives facing landowners and the potential role of policy instruments designed to encourage wetland conservation.

Research examining wetland conservation in the Prairie Pothole Region frequently employs watershed or sub-basin level case studies, as these spatial units provide a natural framework for understanding hydrological processes and landscape-level ecosystem services. Wetlands interact with surrounding land uses through surface and subsurface hydrological connections, meaning that their ecological and economic impacts are often best evaluated at the watershed scale rather than at the level of individual wetlands alone. Previous studies assessing ecosystem service values and policy outcomes have therefore focused on specific drainage basins within the region in order to capture the spatial heterogeneity of wetland distribution, agricultural productivity, and conservation costs (Pattison-Williams et al., 2018; Muhammad et al., 2018). Following this established approach, the present research focuses on a sub-basin of the Assiniboine West Watershed District, allowing for a spatially explicit analysis of wetland drainage costs and conservation benefits within a clearly defined hydrological system.

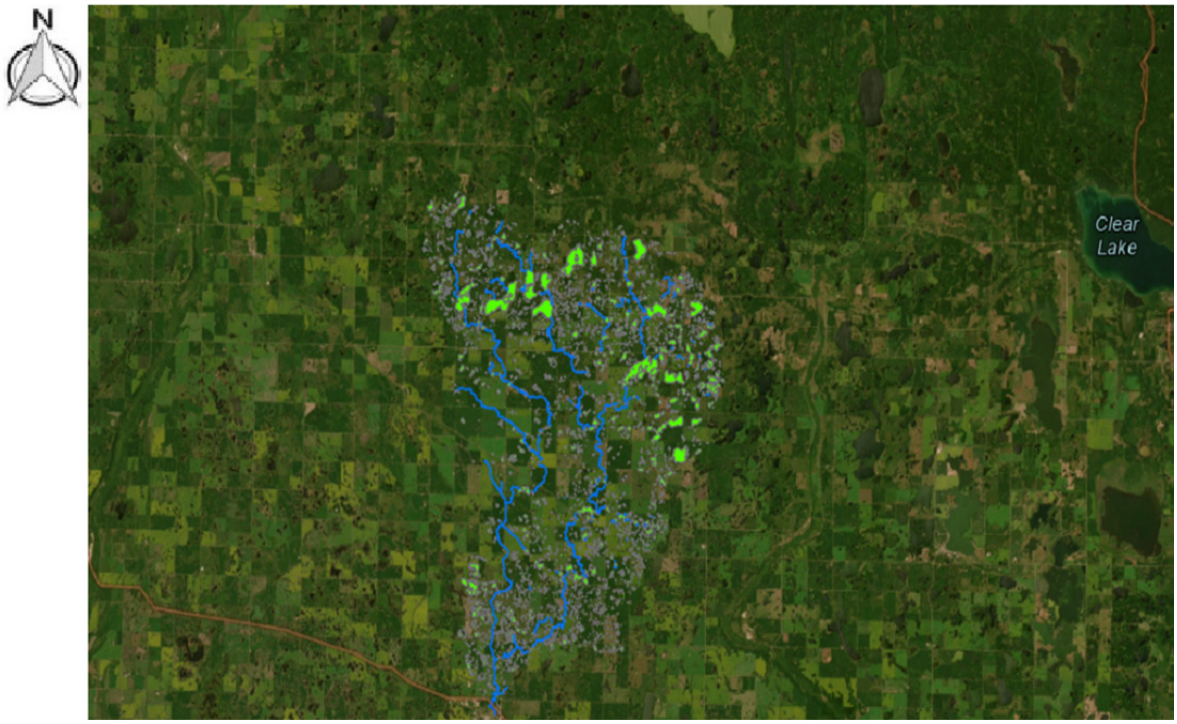


Figure 1. Sub-basin of Assiniboine West Watershed District, Manitoba

Note. bright green = wetlands; blue lines = watercourses.

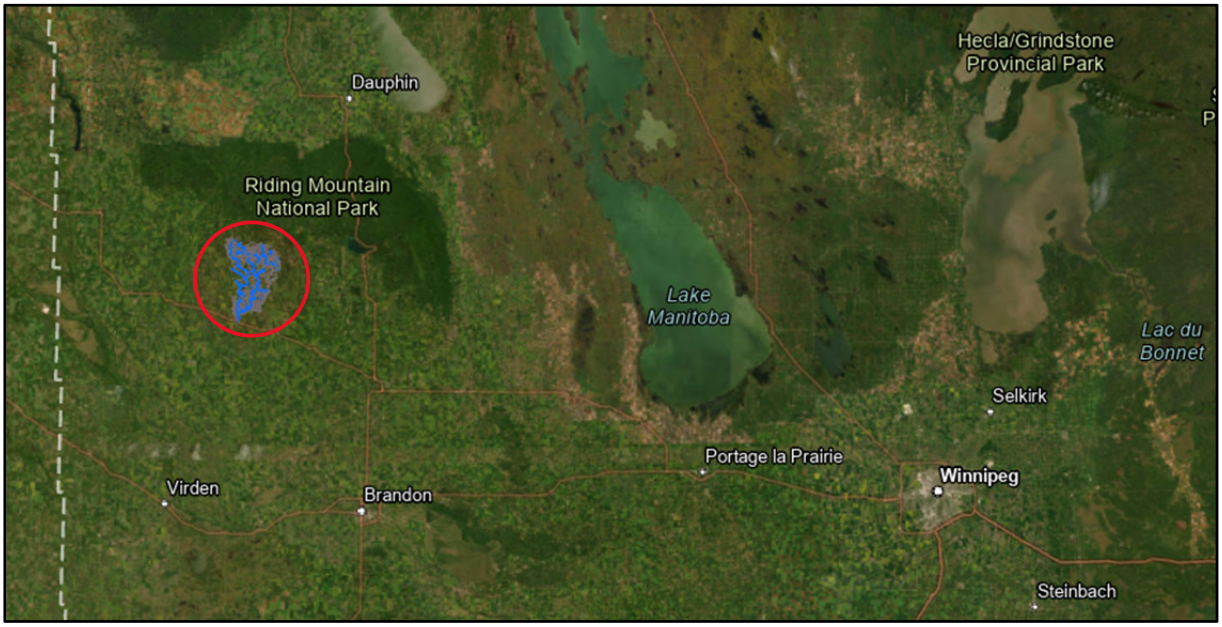


Figure 2. Wider view of Assiniboine West Watershed District sub-basin for geographical context

Note. sub-basin circled in red.

Data from the Manitoba Land Initiative, in partnership with Statistics Canada, supports spatial maps of the wetlands in the study area using ArcGIS mapping software (ESRI Canada, 2023). This allows measurement of the distance from wetlands to a water source to estimate the wetland drainage costs needed for the net present benefit equation. It is assumed that wetlands which are closer to a watercourse have reduced drainage costs given that establishing the necessary drainage infrastructure requires less time and effort compared to a wetland that is further from a watercourse (Asare, Lloyd-Smith and Belcher 2022). Table 1 provides the descriptive statistics of the relevant variables in the wetland dataset used for this research.

Table 1. Variable descriptive statistics, wetland dataset

Variable	Mean	Std. Dev.	Minimum	Maximum
Wetland size (Ha)	8.19	15.30	0.12	337.17
Distance from nearest watercourse (m)	739.01	638.40	0.00	3,853.82
Relative distance (m)	1.00	0.86	0.00	5.22
Drainage cost	$574.20	$271.40	$260.00	$1,898.50

Drainage cost for each wetland in the study is the sum of the 20-year fixed costs associated with maintaining the drained wetland along with the variable costs determined by a wetlands relative distance to the nearest watercourse. To determine the relative distance variable, the proximity of each wetland to the closest watercourse was calculated using the ‘near’ processing tool in ArcGIS Pro mapping software (ESRI Canada, 2023); an average for all wetlands was also computed.

The relative distance variable is the wetland to a watercourse distance divided by average distance of all wetlands to watercourses. This calculation does not consider potentially drained wetlands as watercourses, only the watercourses which already exist. This is a potential limitation of this research as certain wetlands would have shorter relative distances, and therefore lower drainage costs, if the option to connect wetlands via ditches were available. Another limitation regarding the dataset used and the relative distance calculation is the absence of hydrological complexities such as up/downstream flows. This type of information would also be helpful in informing the true costs of draining an individual wetland since the directional flow could have an impact on the size of the ditch. Additionally, this information would help to determine which of the nearest watercourses would be a more realistic option for connecting via ditches.

In this dataset the average distance of all wetlands to watercourses is 739 meters, which results in relative distance values ranging from 0 to 5.22. A distance coefficient is required in order to complete the variable cost component; the rationale for choosing the distance coefficient of 314.2 for this study is that it will ensure that the cost of draining any individual wetland will fall roughly in line with previous estimates of wetland drainage costs. Using this value means that the maximum drainage cost for any one wetland over the entire study area will be roughly $1,900/ha. The fixed cost component of the drainage cost function is assumed to be $260/ha over the 20-year time horizon. This includes all the other costs associated with surface drainage of wetlands, including maintenance and administrative costs. This fixed cost value is based upon the work done by Cortus et al. (2011) on wetland drainage costs in Saskatchewan; this value is appropriate given the similar nature of the study area, as well as the use of a 20-year time horizon and has been adjusted to reflect inflation since that study was conducted. Selection of these values for both the fixed cost and distance coefficient for this research leads to a range of drainage costs for individual wetlands in the study area from $260/ha to $1,900/ha, which are similar to those found in previous work done by Cortus et al. (2011), De Laporte (2014), and Asare, Lloyd-Smith, and Belcher (2022), with adjustments made for inflation. This calibration approach is commonly used in spatial economic models when engineering cost data are unavailable for each individual site, but plausible cost ranges are documented in the literature. Accordingly, Equation (1) gives the wetland drainage cost function

$$DC_i = FC_i + 314.2 \cdot RD_i \quad (1)$$

where, DC_i is the ith wetland's drainage cost; FC_i is the fixed cost; and RD_i is the relative distance to a watercourse of the ith wetland.

A net present benefit (NPB) approach is used for this research to determine the value to private landowners of draining wetlands in the study area; the heterogeneity of the drainage costs of each wetland is incorporated into the NPB estimates. Nine different scenarios were constructed based on a range of potential crop prices and varying levels of soil productivity on the drained wetland. As a result, the NPB equation is calculated under nine different sets of conditions (shown in Table 2) for every wetland in the study area. This determines the net benefits for private landowners to drain a wetland when the ecological benefits to society are not considered.

Table 2. Price and soil productivity levels for low, medium and high scenarios

Scenario	Canola Price (per tonne)	Wheat Price (per tonne)	Soil Productivity (% of maximum yield)
Low	$283.75	$204.50	30
Medium	$340.31	$257.85	50
High	$509.09	$479.69	80

The net present benefit of drainage for this research will be defined as the net present value of crop production returns on the drained wetland, minus wetland drainage cost, over 20 years. Additional analysis was carried out under 30-year and 50-year time horizons to ensure this choice did not significantly impact the results. This analysis demonstrated that the number of wetlands drained, and the associated costs, stayed roughly the same and were not reliant on the value of the time horizon chosen. To carry out this analysis, a crop rotation of spring wheat and canola is used given the prominence of those crops within the study area (Manitoba Government, Ministry of Agriculture, Markets & Statistics, 2023). Data from the Manitoba Government regarding crop prices, production costs, and crop yields are used to estimate the net present benefit of returns from annual cropping; the low, medium, and high price levels were determined by the historical prices of wheat and canola dating back to 1988. The low price level was based on the historically low monthly values seen since 1988 and is included to illustrate a scenario where other economic factors, such as technological change, contribute to a decline in agricultural prices in the future. The medium price level was determined by the monthly average from 1988 to 2022, this illustrates a scenario where agricultural prices remain relatively stable over time. The high price level is an estimate of the future growth in prices which illustrates a scenario where agricultural prices rise significantly over time due to other factors such as climate change, or political unrest for example. Variations in yields after a wetland is drained for the study area were determined via land value assessments from Farm Credit Canada's Farmland Values Report (FCC, 2022); it is assumed that higher land values correlate with higher yield on that land. The lowest land value in the study area is approximately 37% of the highest value, while the average was around 63% of the most valuable land. These ratios were used to determine the differences in the low, medium, and high soil productivity conditions used in the estimations.

The NPB approach utilized in this research does have some limitations since it will not account for certain sociological characteristics that may impact a landowner's decision. It can be argued that previous experience with government conservation programs could affect a landowner's willingness to adopt a new program. Additionally, the unique individual attributes of the landowners (such as age, education level, and risk aversion) could all play a role in determining the value of a wetland. Incorporating this type of decision-making complexity into the model could enhance the analysis at the individual landowner level and is an area that should be explored in subsequent research. However, it is beyond the scope of this research paper.

Another aspect which is left out of this NPB analysis is the impact that neighbouring landowners and the passage of time can have on the decision-making process. In reality, landowners could take a 'wait-and-see' approach over time which would allow them to incorporate new information into their decision-making process. This could include any changes to agricultural commodity prices or other economic factors, as well as the impact from wetland conservation and/or drainage on their neighbour's land. Once again this is an area that is beyond the scope of this research paper but is worth mentioning because it warrants further exploration in future research.

To estimate the present value of crop production over a 20-year period, the annual net-returns must be discounted using a risk-adjusted discount rate; it is therefore also necessary to determine a risk-adjusted discount rate and a time horizon for the study. Capital line market theory (Sharpe et al. 2000), has been used frequently in prior work and is employed in this research; this resulted in use of a risk-adjusted discount rate of 9.6%. Cortus et al. (2011) employed a 10% for Saskatchewan wetland drainage; Koeckhoeven (2008) used 7.5% for cattle and crop production in Alberta; and Asare, Lloyd-Smith and Belcher (2022) utilized a discount rate of 8.3% for wetland drainage in Alberta. The standard deviation of crop returns between 2002 and 2022 was estimated using data from the Manitoba Department of Agriculture (Manitoba Provincial Government, Department of Agriculture, 2023). Expected market returns for this period from the Toronto Stock Exchange came from Yahoo Finance (accessed March 2024). Expected returns from Canadian treasury bills from 2002 through 2022 were estimated using Bank of Canada data (Bank of Canada treasury bill yields, accessed March 2024). Capital market line theory was employed to calculate these values, were then used to compute the risk-adjusted discount rate of 9.6%; this value is in line with discount rates found in the studies cited above.

A sensitivity analysis was conducted with alternative risk-adjusted discount rates to investigate whether this would yield qualitatively different policy recommendations, which would suggest that the outcomes are highly sensitive to assumptions about the future. Alternate estimates were calculated using risk-adjusted discount rates of 6% and 12%. These alternative risk-adjusted discount rates were chosen because they represent the lower and upper bounds of what has been utilized in previous literature which is focussed on long term agricultural returns. Both alternative estimates yielded similar results to the 9.6% rate which was calculated using the capital market line theory approach. For this reason, we moved forward with the analysis for this research using the 9.6% risk-adjusted discount rate.

Equation (2) provides the NPB equation, written as,

$$NPB_i = \begin{bmatrix} \sum_{o=1}^{19} (a_i) \times (SoilProd \times Price_o^C \times Yield_o^C - Cost_o^C) \times (1+r)^o \\ + \sum_{e=2}^{18} (a_i) \times (SoilProd \times Price_e^{sw} \times Yield_e^{sw} - Cost_e^{sw}) \times (1+r)^e - (a_i \times DC_i) \end{bmatrix} / a_i \quad (2)$$

where, NPB_i is the net present benefit of the ith wetland drained (in dollars per hectare); a_i is the area of the ith wetland drained; *SoilProd* is the soil productivity condition; *Price* and *Yield* give the price and yield of canola and wheat as noted in the equation; r is the private risk-adjusted discount rate of 9.6%; o is for odd years (when canola is planted in the canola/wheat rotation) and e is for even years (when wheat is planted); *Cost* is the specific production cost associated with each crop, and DC_i is the drainage cost per hectare of drained land. This follows the approach of Asare, Lloyd-Smith, and Belcher (2022), whose research was carried out for a region of Alberta, Canada.

Estimation of the economic benefits to society of conserving wetlands will help inform ideal policy targeting for this region. Previous work has used wetland classification data to assign a beneficial wetland ratio to each quarter-section in their study area; this allowed researchers to account for heterogeneity in wetland ecosystem benefits in addition to the heterogeneity of wetland drainage costs. Unfortunately, these data are not accessible for the province of Manitoba; instead, this research uses an approach informed by the work of De Laporte (2014). However, the technique with which the data is applied in this research makes use of findings from the hydrology and earth sciences fields. Researchers in those fields have linked the value of ecosystem benefits that wetlands provide to both the overall size of a wetland, as well as how isolated an individual wetland is from other

wetlands within the same watershed (Evenson et al., 2015; Ahlen et al., 2020). The monetary ecosystem benefits data are derived from the estimates of the value of the natural capital in the Upper Assiniboine River Watershed completed by Belcher, Edwards, and Gray (2001), which have been updated to reflect inflation (Note 1). Values include savings from government payments and crop insurance premiums foregone as a result of conservation activities, water quality improvements resulting from decreased sediment, reduced erosion from wind, improved presence of wildlife for both viewing and potentially hunting; benefits associated with water-based recreation, and environmental benefits including carbon sequestration and reductions in greenhouse gas emissions. While it is worth acknowledging that these ecosystem benefits were calculated over 20 years ago, after being adjusted for inflation, they remain closely in line with more recent estimates which were completed on watersheds in different geographic regions (Gustavson & Kennedy, 2010; Richardson, Liu, & Eggleton, 2022; Li et al., 2020). Since this data comes from a neighbouring watershed to the one analyzed in this research, it provides the most reliable estimate possible without having access to the granular mapping data.

To construct a supply curve for wetland conservation, all wetlands are ranked from lowest-to-highest-cost. Wetlands with negative NPBs are given an area of 0 ha and an NPB of $0/ha. The size of a wetland and its relative distance to the nearest watercourse were used to determine the proper distribution of ecosystem benefits based on previous research done by Evenson et al. (2015) and Ahlen et al. (2020). Wetland ecosystem benefits vary across landscapes depending on physical and spatial characteristics of individual wetlands. Two factors frequently identified in the literature as important determinants of wetland ecosystem function are wetland size and spatial isolation. Larger wetlands generally provide greater hydrological storage capacity, improved nutrient retention, and enhanced wildlife habitat compared with smaller wetlands (Gleason, Euliss, & Tangen, 2011; Johnston & McIntyre, 2019). Similarly, wetlands that are more connected to surrounding wetlands or watercourses tend to support greater ecological function and biodiversity than highly isolated wetlands because connectivity facilitates hydrological exchange and species movement across the landscape (Vanderhoof, Alexander, & Todd, 2016). Following this literature, wetlands in this study were categorized into high, medium, and low ecosystem benefit levels based on combinations of wetland size and relative isolation. These categories are intended to capture relative differences in ecosystem service provision within the study area rather than represent absolute ecological thresholds, which are highly context-specific across prairie landscapes. Because ecological thresholds for ecosystem service provision vary across prairie landscapes, benefit categories were defined using relative thresholds based on the distribution of wetland size and isolation within the study area, allowing wetlands to be ranked according to their expected ecosystem service contribution.

For this research, the following criteria was used to assign a level of ecosystem benefits provided by each wetland: if the area of a wetland is greater than 30ha (two times the standard deviation), it is considered a high benefit wetland; if the relative distance of a wetland is greater than 3, and the area is greater than 0.75ha, it is also considered a high benefit wetland; if the area of a wetland is less than 8.2 ha (the average size of all wetlands in the study area) and has a relative distance less than 3, then it is considered a low benefit wetland. Any remaining wetlands are considered medium benefit wetlands. This results in 209 high benefit wetlands which represents 37% of the total area of wetlands in the study area, 517 medium benefit wetlands which represents 36% of the total area of wetlands, and 1,823 low benefit wetlands which account for the final 27% of the total area of wetlands.

After classifying each of the wetlands in the study area, the total costs and the cost per hectare of conserving wetlands are estimated under various policy targeting scenarios. These costs, as well as the total area of wetlands conserved, are compared to illustrate the trade-offs between the different policies. The policy targeting scenarios are: (a) high-benefit targeting, which only conserves wetlands which have been classified as having high ecosystem benefit values; (b) low-benefit targeting, which only conserves wetlands with the lowest relative ecosystem benefit values; (c) medium-benefit targeting, which only conserves wetlands with medium ecosystem benefit values; and (d) a strategy which targets the wetlands that provide the lowest amount of net present benefit for private landowners.

4. Results

Table 3 presents the amount of wetland drainage occurring under nine different scenarios, based on combinations of crop prices and varying levels of soil productivity. For all nine scenarios, the NPB of each wetland was estimated using Equation (2) above to determine the number of wetlands that would be drained by private landowners with no regulations in place to restrict drainage activity and the ecosystem benefits generated for society are not included. Results suggest that the number of wetlands drained will increase as prices for agricultural commodities do. The same is true for the perceived productivity of the soil on the parcel of land where the wetland is drained; according to the NPB estimates calculated for this research, the impact of soil productivity on the number of wetlands drained is greater than the effect of agricultural commodity prices when

the soil productivity is assumed to be high. This impact is less significant when low soil productivity is assumed; for example, in the scenarios with the highest possible soil productivity, the NPB of draining each wetland was positive regardless of the price level. This suggests it is beneficial to private landowners to drain all the wetlands in the study area under these soil productivity conditions at any price level. In the low soil productivity conditions, it is not economically beneficial for private landowners to drain any of the wetlands at the low and medium price levels; however, under the high price/low soil productivity scenario, all but 33 wetlands are drained. This indicates that price does have an impact on a landowner's decision whether to drain a wetland or not, and this impact is not entirely dependent on whether the soil is highly productive. This is congruent with the findings of previous work in this area which has found that price levels of agricultural commodities are highly indicative of the number of wetlands that will be drained in an agricultural region (De Laporte, 2014; Asare, Lloyd-Smith, & Belcher, 2022; Cortus et al., 2011; Koeckhoven, 2008).

Table 3. Wetland drainage results without ecosystem benefits considered

Soil productivity condition	Price level	# of wetlands drained	% of wetlands drained	Average	Std. Dev.	Maximum
				---------- NPB/ha ----------		
Low	Low	0	0	$0.00	$0.00	$0.00
	Medium	0	0	$0.00	$0.00	$0.00
	High	2,516	98.7	$416.27	$543.40	$6,902.60
Medium	Low	2,192	85.9	$147.86	$218.39	$3,098.28
	Medium	2,510	98.5	$401.48	$525.64	$6,708.22
	High	2,549	100.0	$1,473.91	$1,855.00	$19,891.97
High	Low	2,549	100.0	$867.93	$1,089.34	$12,506.03
	Medium	2,549	100.0	$1,341.82	$1,686.48	$18,281.94
	High	2,549	100.0	$3,072.49	$3,914.34	$44,199.91

Exploring the mean NPB under each scenario with no regulations in place also provides perspective on how agricultural commodities and soil productivity affect the overall land value, which in turn determines the financial viability of draining the wetland. Under the low soil productivity conditions, results suggest it is economically beneficial to drain wetlands only at the highest crop price level. This resulted in a mean NPB per hectare of $416.27/ha, and a standard deviation of $543.40/ha. Under the medium soil productivity conditions, mean NPB per hectare ranges from $147.86/ha for the low price level to $1,473.91/ha for the high price level; standard deviations range from $218.39/ha to $1,855/ha. Under high soil productivity conditions, the mean NPB per hectare range spikes to between $867.93/ha and $3,072.49/ha, with standard deviations between $1,089.34/ha and $3,914.34/ha. Asare, Lloyd-Smith, and Belcher (2022) found a mean net present benefit of $2,404/ha and a standard deviation of $2,796/ha; this result is most similar to the high price/high soil productivity scenario in this research paper.

As part of the analysis, an average annualized wetland conservation cost was calculated for each soil productivity scenario which can be compared to the estimates generated by previous research conducted on similar agricultural regions. This research found an average annualized wetland conservation cost of $107/ha under low soil productivity conditions, $165/ha under medium soil productivity conditions, and $435/ha under high soil productivity conditions. Asare, Lloyd-Smith, and Belcher (2022) calculated a value of $187/ha for a sub-basin within Alberta's Prairie Pothole Region, roughly in line with the medium soil productivity scenario in this research. Hill et al. (2011) computed a value of $341/ha as part of their research into restoration costs for drained wetlands in Eastern Saskatchewan's Assiniboine River Watershed, which also fits in the range found in this research paper. However, Cortus et al. (2011) calculated the average annual net benefit from draining wetlands to be between $55/ha and $66.75/ha (in 2023 dollars) for a representative farm in east central Saskatchewan, significantly less than the estimates generated by this research; disparities in the estimates may be attributable to variations in the crop rotation assumed. Additionally, Cortus et al. (2011) used a representative farm based on the characteristics of a specific rural municipality, while this research paper allowed for a broader range of wetland conservation costs through the estimation of nine different scenarios regarding crop prices and soil productivity. Work by Lawley and Towe (2014) also calculated a reduction of $53/ha (in 2023 dollars) for wetlands under permanent easement programs in Saskatchewan. Again, there were significant differences in the manner of the research conducted as those authors used a reverse auction approach and focused solely on

easements. Overall, results indicate the estimates made in this research may be on the high side for the region but are in line with the findings of past research. There are limitations to relying on averages to inform wetland conservation policy at any rate, which is demonstrated by spatial differences in wetland conservation costs observed in this analysis.

Table 4 presents the number of wetlands that should be conserved based on whether the ecosystem benefits of a particular wetland outweigh the NPB to the private landowner of draining that wetland. Due to the difficulty in determining a monetary value for the various ecosystem benefits wetlands provide for society, three different levels of monetary value are considered under each of the nine price/soil productivity scenarios estimated. The monetary values used here correspond with the results found by Belcher, Edwards, and Gray (2001), adjusted for inflation. This research estimated the overall societal benefits of a conservation program in the Upper Assiniboine River Basin by examining the effects that wetlands have on a variety of different factors that impact an ecosystem. High, low and best (medium) estimates are provided in order to be congruent with the range of feasible outcomes and the inherent uncertainty implicit in estimation of costs and benefits. The high estimate is intended to provide an upper bound for the benefits (and lower bound for the costs) while the low estimate provides a lower bound for the benefits (and upper bound for costs). Every individual wetland was then categorized as a low, medium, or high benefit wetland. This was based on the criteria outlined in the previous section which is informed by research from the hydrology field that has linked wetland size and isolation to the ecosystem benefits they provide. This provides the range of potential outcomes for the future of wetland drainage in this sub-basin of the Assiniboine West Watershed District in Manitoba.

Table 4. Wetland drainage results with ecosystem benefits considered

Soil productivity condition	Price level	# of wetlands conserved	% of wetlands conserved	Cost per hectare	Total cost (millions)
Low	Low	0	0	$0.00	$0.00
	Medium	0	0	$0.00	$0.00
	High	1,736	94.8	$79.09	$1.565
Medium	Low	2,376	99.4	$28.11	$0.584
	Medium	1,772	95.1	$77.30	$1.535
	High	730	71.4	$149.53	$2.310
High	Low	1,144	85.4	$120.10	$2.142
	Medium	789	76.0	$143.37	$2.274
	High	677	72.5	$294.99	$4.469

In low soil productivity scenarios, private landowners do not have much economic incentive to drain wetlands in the first place, even before considering the benefits they provide to society. As stated above, none of the wetlands are economically beneficial to drain at the low and medium price levels under a low soil productivity scenario. At high price levels and low soil productivity 1,736 wetlands would still be conserved. These wetlands would represent roughly 95% of the total area of wetlands in the study area. These results indicate that in a scenario where the perceived soil productivity of wetlands is relatively low, the ecosystem benefits provided by wetlands to society far outweigh the benefits private landowners would generate from draining the wetlands. In the low soil productivity environment with high prices, a policy which paid private landowners to conserve all wetlands on their property would cost $79.09/ha, and result in a total cost of $1.56 million.

The outcomes under the medium soil productivity scenario illustrate the importance of considering the ecosystem value wetlands provide to society when deciding whether to drain or conserve a wetland. With no regulations in place to force private landowners to consider ecosystem benefits, it is socially beneficial to drain all wetlands under high price levels. Under medium price levels, all but 39 of the wetlands are drained, and under low price levels 86% of all wetlands are still drained. When the benefits wetlands provide for local ecosystems are considered in the calculation, it becomes clear that leaving these decisions solely in the hands of private landowners will result in an underinvestment of environmental goods. According to the analysis conducted, under low price levels and medium soil productivity, 93% of all wetlands should be conserved instead of only 14% when ecosystem benefits are not considered. The results of the estimates of both the medium and high price levels under medium soil productivity illustrate the importance of the price of agricultural commodities in both determining which wetlands are worth conserving and determining how much it would cost to undertake the conservation through payments to private landowners. The analysis shows that under medium price/medium soil

productivity, 70% of all wetlands should be conserved compared to less than 2% when ecosystem benefits are not considered.. The cost to conserve all wetlands in this scenario that have higher social net present benefit under a hypothetical conservation policy would be $1.53 million; this works out to $77.30/ha. The analysis of the high price/medium soil productivity scenario reveals that only 29% of wetlands should be conserved. When ecosystem benefits are not considered in this scenario, the results indicated that all of the study area's wetlands should be drained. The cost of conserving the wetlands in this scenario is $2.31 million. This translates to $149.53/ha, which is 93% higher than under the medium price scenario.

The results under high soil productivity conditions further demonstrate the importance of the price of agricultural commodities in determining the overall societal value of a wetland because the disparity between the number of wetlands drained when ecosystem benefits are not considered versus when they are is even more extreme. When ecosystem benefits are not considered, this scenario reveals that 100% of the wetlands in the study area should be drained and converted to agricultural production regardless of the price levels. Under low prices/high soil productivity, with the ecosystem benefits considered, the number of wetlands conserved is 45% which costs $2.14 million. This works out to $120.10/ha. In the medium price/high soil productivity scenario, 31% (789/2,549) of wetlands are conserved at a cost of $143.37 per hectare, totalling $2.274 million. In the high price/high soil productivity scenario, 27% of the total wetlands are conserved and the total cost is $4.47 million. This works out to $294.99/ha.

De Laporte (2014) found that under historically high prices, 15% of the total wetlands in his study area should be conserved based on their NPB. He also found that under a historically low-price situation, almost all the wetlands were not economically beneficial to drain irrespective of the level of ecosystem benefits used. Both results are very similar to the estimates found in this research. However, under De Laporte's medium price scenario, only 5% of the total wetlands would be conserved, far less than what is conserved under a similar scenario in the present study. This may be primarily due to differences in the crop rotation, prices, and yields used to calculate the net present benefit, because the same ecosystem benefit values were applied in the analysis done by Belcher, Edwards, and Gray (2001). Asare, Lloyd-Smith, and Belcher (2022) found that 38% of the wetlands in their study area had negative net present benefits; this result tracks closely with the range of wetlands conserved under both the high price/medium soil productivity, and the high price/high soil productivity scenarios in this research. Cortus et al. (2011) found a similar result for their representative farm in east central Saskatchewan as 46% to 51% of wetlands in their study area were drained depending on the size of the farm.

Figure 3 displays the supply curves of the low, medium, and high soil productivity scenarios. This demonstrates the impact that soil productivity has on the total hectares of wetlands that are conserved. The supply curve's upward slope shows the positive effects of wetland conservation cost upon total potentially conserved wetland area, all else being equal. There is also an increase in the slope of the curves which demonstrates that the level of elasticity reduces along the curves. On the flatter part of the curves, the number of wetlands conserved increases significantly in response to relatively small increases in conservation costs imposed on landowners. The final stage of all three curves is highly inelastic, meaning significant increases in the conservation costs imposed on landowners are required in order to make conserving these wetlands economically viable.

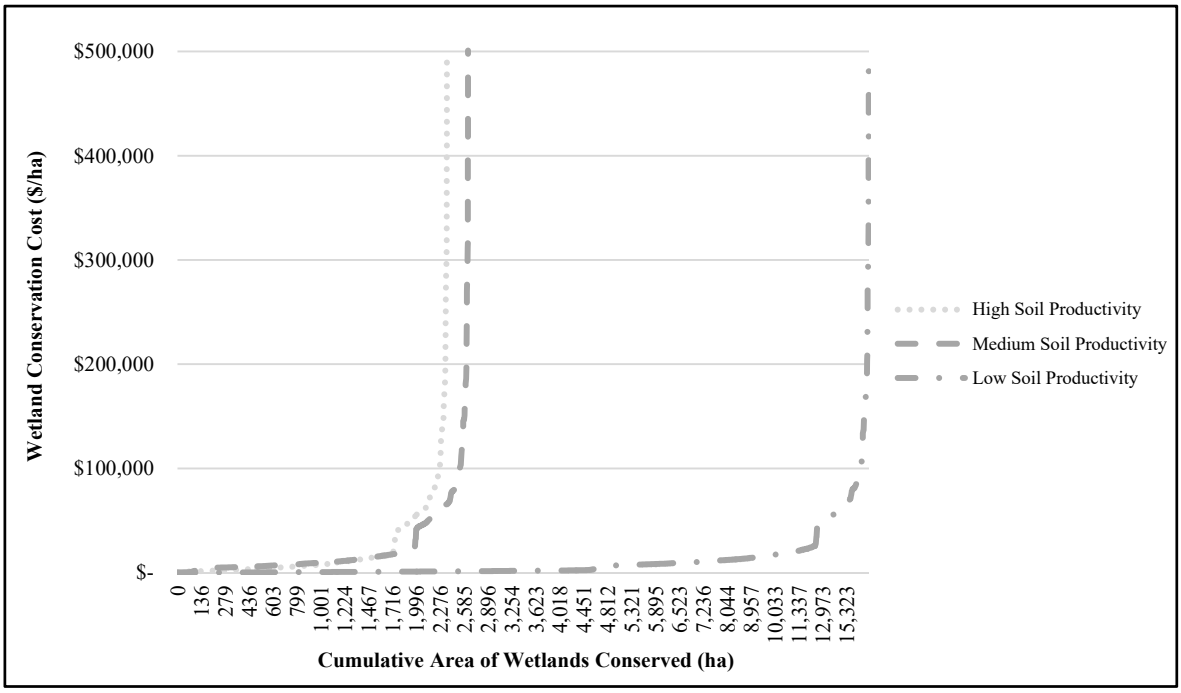


Figure 3. Supply curves for low, medium & high soil productivity scenarios

Figure 4 then displays the supply curves of the low, medium, and high price scenarios when medium soil productivity is assumed for all wetlands. This chart displays the impact that agricultural commodity prices have on the total hectares of wetlands that are conserved.

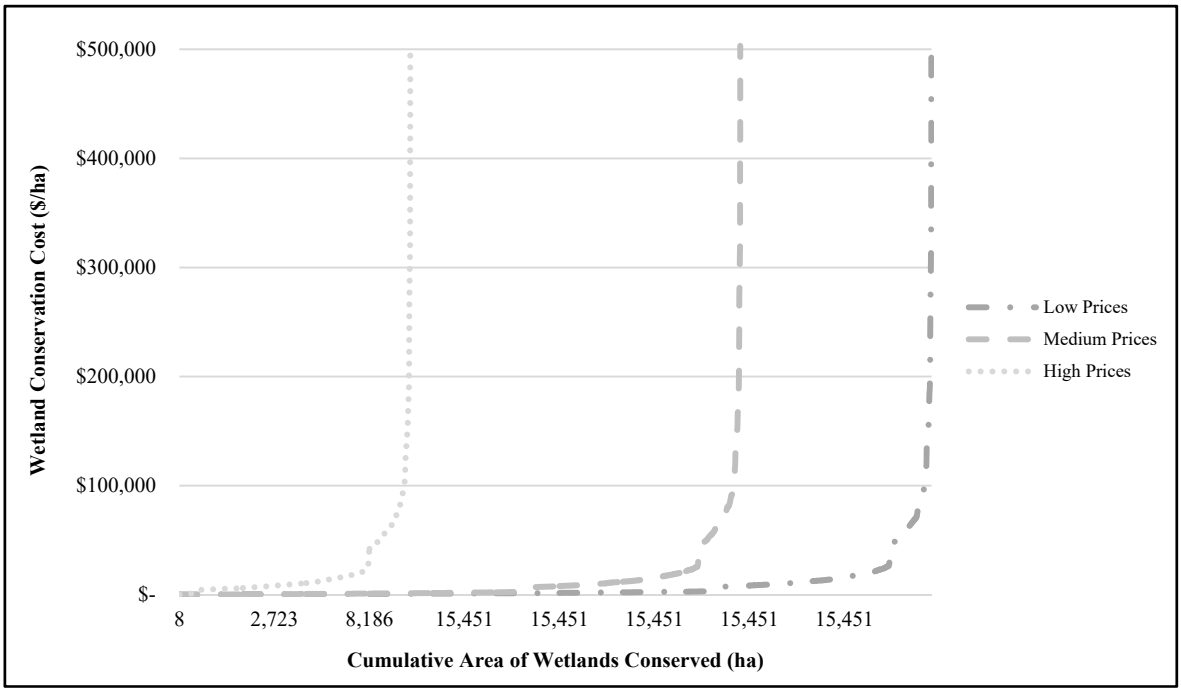


Figure 4. Supply curves for low, medium, & high price scenarios under medium soil productivity

The final aspect of the analysis in this research facilitates evaluation of various policy targeting scenarios so that a better understanding can be gained of the costs involved when policymakers construct policies with specific

environmental goals in mind. This analysis was done under the high soil productivity scenario, for each of the three price levels. A policy designed to target only wetlands with high ecosystem values would cost between $333,674 ($43.40/ha) and $1.31 million ($170.92/ha) and would protect around 37% of the total area of wetlands. A policy designed to target only the medium benefit wetlands would cost between $1.01 million ($133.66/ha) and $3.44 million ($455.02/ha) and would save roughly the same total area as the policy targeting only high benefit wetlands. This is clearly a less efficient policy outcome since both the overall cost and the cost/ha are higher, while the ecosystem benefits realized will be smaller. If instead policymakers target only low benefit ecosystem value wetlands, the total cost would be between $3.51 million ($620.68/ha) and $12.06 million ($2,135/ha). This policy outcome would be less efficient than both the high and medium targeting policies due to the higher costs and lower level of ecosystem benefits retained.

To demonstrate the importance of considering the heterogeneity of ecosystem benefits when designing a policy which will pay landowners to compensate them for not draining wetlands, the cost of a policy which was based solely on targeting wetlands which provided the lowest net present benefit to private landowners, rather than using the net present benefit to society, was calculated. This was once again done for the high soil productivity scenarios. In this hypothetical, policymakers are unaware of the exact ecosystem benefits that each wetland provides, and they decide to save one third of all the wetlands in the sub basin of the watershed based solely on which are the least valuable to private landowners as agricultural producing land. Implementing such a policy would cost between $1.36 million ($217.29/ha) and $5.34 million ($853.65/ha), depending on the price levels experienced over time, and it would result in roughly 30% of the total hectares of wetlands being conserved. This result is clearly less efficient than the outcome that results from a policy which aims to conserve only the high benefit wetlands since it is both more expensive and results in less hectares of wetlands being retained. Additionally, the policy targeting high benefit wetlands is clearly a better result from an ecological perspective since we know that the cumulative ecosystem benefits of that policy are greater than the benefits from the policy which is based solely on private net present benefit.

5. Summary and Conclusions

This research estimated the economic benefits of wetland drainage in a sub-basin of the Assiniboine West Watershed District in Manitoba for private landowners using a wetland conservation cost model with spatial dimensions. These estimates facilitated the calculation of wetland conservation supply curves with declining elasticity as more wetlands are conserved, consistent with the findings of previous work. This research found that as agricultural commodity prices increase, so does the number of wetlands drained. It further discovered that soil productivity has a more prominent effect than crop price on a landowner's decision to drain a wetland. Under low soil productivity conditions, it is not economically beneficial for private landowners to drain any wetlands at low and medium price levels. Under the highest possible soil productivity conditions, the net present benefit of draining every wetland was positive regardless of the price level. An average annualized wetland conservation cost was also calculated for each soil productivity scenario and was found to be $107/ha under low soil productivity, $165/ha under medium soil productivity, and $435/ha under high soil productivity conditions. Past literature has produced results ranging from $55/ha to $341/ha (Asare, Lloyd-Smith, & Belcher, 2022; Hill et al., 2011; Cortus et al., 2011; Lawley & Towe, 2014).

Private economic benefits of wetland drainage were then compared to estimates for the public value of wetland retention in a cost-benefit analysis. The ecosystem benefits, which represent the public value of wetland retention, were assigned to the wetlands in the study area based on the research of Evenson et al. (2015) and Ahlen et al. (2020). The monetary value of these ecosystem benefits was derived from Belcher, Edwards, and Gray (2001). This analysis highlights the importance of taking the public benefits of wetlands into consideration when designing conservation policies. Without those benefits included in the analysis, market failure can occur where the number of wetlands preserved is below the optimal level due to the actions of private landowners who are not incentivized to consider the benefits of wetland retention to the public.

The results of both the high and medium soil productivity scenarios illustrate the importance of how wetlands are valued in this type of analysis since the number of hectares of wetlands conserved, and the associated costs, varies significantly when the ecosystem benefits to society are considered in the estimation of the wetlands total value. Under medium prices/medium soil productivity, 70% of all wetlands would be conserved when ecosystem benefits are considered compared to only 2% when the decision is based solely on the private landowners net present benefit. When high prices are combined with medium soil productivity the result is 27% of all wetlands being conserved and the cost to conserve them is 93% higher than the medium price scenario. These results illustrate the importance of agricultural commodity prices in determining the costs of a policy as well as the number of wetlands that are conserved.

Under low price/high soil productivity, the number of wetlands conserved is 45% of the total wetlands when ecosystem benefits are considered, and the cost of this policy would be $2.14 million ($120.10/ha). In the medium price/high soil productivity scenario, the number of wetlands conserved is 31% and the total conservation cost is $2.27 million ($143.37/ha). These results demonstrate the importance of soil productivity in determining the success of wetland retention policies because the number of wetlands conserved, as well as the total hectares these wetlands represent, falls significantly in the high soil productivity scenario when compared to the medium soil productivity scenarios.

Results of the cost-benefit analysis provide a basis for consideration of potential wetland conservation policies intended to promote growth in the agriculture industry while minimizing the negative environmental impact on wetlands. Four different policy targeting scenarios were assessed, under the high prices/high soil productivity scenario, to provide estimates of the costs and the percentage of wetlands that would be conserved. If a policy is designed to target only wetlands with high ecosystem values, it would cost between $333,674 ($43.40) and $1.31 million ($170.92/ha) depending on the price levels; this policy would protect around 37% of the total area of wetlands. The second hypothetical policy would target only medium benefit wetlands. This would cost between $1.01 million ($133.66/ha) and $3.44 million ($455.02/ha) and protect roughly the same number of wetlands as the high benefit targeting policy. A policy targeting only low ecosystem value wetlands would cost between $3.51 million ($620.68/ha) and $12.06 million ($2,135.00/ha), and conserve only 27% of all wetlands in the study area. It is clear from these results that policymakers should construct policies which attempt to conserve the highest ecosystem benefit wetlands. More research is needed in this area to ensure that policymakers are equipped with the proper data to construct such a policy.

The final policy targeting scenario considered a situation where the policymakers cannot determine the ecosystem benefits of every individual wetland and instead decide which to conserve based on their net present benefit to the private landowners of each wetland. This policy is designed to target one-third of all the wetlands in the study area and results in roughly 30% of the total hectares of all wetlands being conserved at a cost between $1.36 million ($217.29/ha) and $5.34 million ($853.65/ha). This outcome is less efficient economically compared to the high benefit targeting policy. It also leads to an ecologically worse outcome because less total wetlands are conserved and some of the most ecologically valuable wetlands will have been drained. These results underscore the importance of considering the societal benefits of wetlands and suggest that policy design should take into account both environmental and economic impacts. Using simplified criteria based only on the costs of draining and how that relates to the net present benefit of the land to private landowners can lead to less optimal outcomes for society.

Future research in this field could help to construct a clearer view of how to value wetlands from the perspective of both the private landowner and all of society. As mapping and data gathering techniques continue to improve, the current gaps in this area of research will be more easily addressed. The ability to introduce heterogeneity in wetland ecosystem benefits into a model such as the one employed in this paper would help policymakers to better determine the optimal wetland retention policy for the long-term future of wetlands and the watersheds they are a part of, as well as for the economic growth and development of the region. It could also provide individual private landowners with the ability to assess their own land in a manner that is both economically and ecologically efficient.

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Notes

Note 1. These watershed data were chosen because it is one of the few watersheds in this region of Canada for which wetland valuation estimates have been made, and it is also a neighbouring watershed of the study area for this research.

Acknowledgments

The helpful comments of Chad Lawley and Greg Mason on earlier versions of this work are gratefully acknowledged.

Authors Contributions

Shayne St. Denis and Jared Carlberg were responsible for study conceptualization. Shayne St. Denis was responsible for study design, analysis and lead authoring the first draft. Jared Carlberg was responsible for co-authoring the first draft, finalization of the manuscript, and subsequent revisions through the publication process. Both authors read and approved the final manuscript.

Funding

Support for the publication charge for this work was provided through the Solomon Sinclair Farm Management Institute at the University of Manitoba.

Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Informed Consent

Obtained.

Ethics Approval

The Publication Ethics Committee of the Canadian Center of Science and Education.

The journal's policies adhere to the Core Practices established by the Committee on Publication Ethics (COPE).

Provenance and Peer Review

Not commissioned; externally double-blind peer reviewed.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Data Sharing Statement

No additional data are available.

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